Let X be a set. A topology on X is a collection $\mathcal{T}$ of open subsets of X called open subsets, satisfying

i) X and $\emptyset$ are open
ii) The union of any family of open subsets is open
iii) The intersection of any finite family of open subsets is open

A pair $(X, \mathcal{T})$ of a set and a topology on the set is a topological space.

- A neighborhood of $p \in X$ is an open subset containing p.
- S is said to be closed if $X \setminus S$ is open

- A collection $\mathcal{B}$ of open subsets of X is a basis for the topology of X if every open subset of X is the union of some collection of elements of $\mathcal{B}$

Equivalently if X is a set and $\mathcal{B}$ is a collection of subsets of X, $\mathcal{B}$ is a basis if

i) $X = \bigcup_{B \in \mathcal{B}} B$
ii) if $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$ then $\exists B_3 \in \mathcal{B}$ s.t. $x \in B_3 \subseteq B_1 \cap B_2$

A set is countable if $\exists$ a bijection between it and a subset of $\mathbb{N}$

A topological space is second-countable if there is a countable basis for its topology.

Homeomorphism

A continuous bijective map $F: X \to Y$ with a continuous inverse F^{-1}

Continuous

A map $F: X \to Y$ is continuous if $\forall$ open subset $U \subseteq Y$ the preimage $F^{-1}(U)$ is open in X.

A topological n-manifold M is a topological space with the following properties
- M is a Hausdorff space
  - $\forall$ pairs of distinct points $p, q \in M$ $\exists$ disjoint open subsets $U, V \subseteq M$ s.t. $p \in U$ and $q \in V$
- M is second-countable
  - $\exists$ a countable basis for the topology of M
- M is locally Euclidean of dimension n
  - $\forall p \in M$ $\exists$ neighborhood around p that is homeomorphic to an open subset of $\mathbb{R}^n$
  - Given coordinate charts (U, φ) $U \subseteq M$, $\varphi: U \to \hat{U} \subset \mathbb{R}^n$, φ homeomorphism

Ex: $\mathbb{R}^n$, n-sphere, $M_n(\mathbb{R})$, torus, Graph of continuous function, $SO(n, \mathbb{R})$, $Sym(n, \mathbb{R})$

Smooth manifolds:

If $U \subseteq \mathbb{R}^n$, $V \subseteq \mathbb{R}^m$ $F: U \to V$ is smooth if it has continuous partial derivatives of all orders.

If F is a bijection, and F^{-1} is smooth, F is a diffeomorphism.

Given two charts (U, φ) and (V, ψ) are smoothly compatible if $U \cap V = \emptyset$ or $\psi \circ \varphi^{-1}$ is a diffeomorphism.

Atlas are a set of charts that cover M

A smooth atlas is an atlas where all pairs of charts are smooth/compatible.

A smooth atlas $\mathcal{A}$ is maximal if it is not contained in any other smooth atlas

A smooth manifold $(M, \mathcal{A})$ where M is a topological manifold and $\mathcal{A}$ is a smooth atlas.

A smooth Manifold M is a topological space s.t. $U_p \subseteq M$ s.t. U_p is an open neighborhood of $p \in M$
$\exists$ smooth function $\varphi: U_p \to \hat{U}_p \subseteq \mathbb{R}^n$ and if $U, V \subseteq M$ $\varphi_U^{-1} \varphi_V : V \to U$ is smooth

Thm: Regular-Level-Set Theorem

X, Y smooth manifolds

let $f: X \to Y$ be smooth

if dF is surjective along $F^{-1}(0)$, this level set is a smooth submanifold of X with tangent space $\ker dF$ with degree $\deg(X) - \deg(Y)$

Consider $F: M_3(\mathbb{R}) \to Sym_3(\mathbb{R})$

$F(Q) = Q^T Q - I$

$F^{-1}(0) = \{Q : Q^T Q - I = 0\}$

$= \{Q : Q^T = Q^{-1}\}$

$= O(3)$

$dF_Q(V) = \frac{d}{dt} F(Q + tV)$

$= Q^T V + V^T Q \in Sym_3(\mathbb{R})$

$S \in Sym_3(\mathbb{R})$ is mapped onto under dF_Q for $Q \in O(3)$ by $\frac{QS}{2}$

$\Rightarrow O(3)$ is a 3-dim submanifold of $O(3)$

Note: $SO(3)$ is an open subset of $O(3)$

Tangent Space of SO_3 at Q, $T_Q SO(3)$, $Q \in O(3)$

$= \ker dF_Q = \{Q^T V + V Q = 0\}$

$= \{QV : V^T = -V\}$

Define $\mathfrak{so}(3) := T_I SO(3) = \{A : A^T = -A\}$

A lie group is a smooth manifold G (without boundary)
that is also a group with maps $m: G \times G \to G$ and $i: G \to G$

$$m(g,h) = gh \quad \text{and} \quad i(g) = g^{-1}$$

which are both smooth

$SO(3)$ is a lie group under matrix multiplication and matrix inversion

Define $\exp(tA) = \sum_{k=0}^{\infty} \frac{t^k A^k}{k!}$

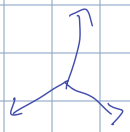
For $A \in SO(3)$ $\quad \exp(tA)^T = \sum_{k=0}^{\infty} \frac{t^k (A^T)^k}{k!} = \sum_{k=0}^{\infty} \frac{(-t)^k A^k}{k!} = \exp(-tA)$

$$\Rightarrow \exp(tA)\exp(tA)^T = \exp(tA)\exp(-tA) = \exp(0) = \underline{I}$$

Define map $\mathbb{R}^3 \to SO(3)$

$$\omega \mapsto \hat{\omega}$$

$$(x,y,z) \mapsto \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix} \quad \text{Note } \hat{w}v = w \times v$$



For a unit vector u.

$$\exp(t\hat{u}) = I + (\sin t)\hat{u} + (1-\cos t)(\hat{u})^2$$

This is rotation a vector $v \in \mathbb{R}^3$ by t radians along the u axis

A tiny rotation is $I + \Delta t \hat{u}$

A huge rotation can be split into n small ones by

$$\left(I + \frac{t}{n} A\right)^n$$

Taking $\Delta t \to dt$ $\quad n \to \infty \Rightarrow \lim_{n\to\infty} \left(I + \frac{t}{n}A\right)^n = e^{tA}$ (Not really but the intuition is there